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1) Point charge 1mc and -2mc located at (3, 2, -1) and (-1, -1, 4) respectively Calculate the electric force on a 10nc charge located at (0, 3, 1) and the electric field intensity at that point?

Sol: Given Data

charge $q_1 = 1\times10^{-3} \rightarrow p(3,2,-1)$ Test charge $q_0 = 10\times10^{-9}c \rightarrow p(0,3,1)$

$q_2 = -2\times10^{-3} \rightarrow p(-1,-1,4)$

We know that Electric field intensity $\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{a}_r \rightarrow (1)$

Electric force on a charge $\vec{F} = q_0\vec{E} \rightarrow (2)$

1) Electric field p due to q_1:

$\vec{r}_1 = (0-3)\hat{i} + (3-2)\hat{j} + (1+1)\hat{k}$

$= -3\hat{i} + \hat{j} + 2\hat{k}$

$r_1 = \sqrt{14}$

$\hat{r}_1 = (-3\hat{i} + \hat{j} + 2\hat{k})/\sqrt{14}$

$\vec{E} = 9\times10^9 \times \frac{1\times10^{-3}}{14}\left(\frac{-3\hat{i}+\hat{j}+2\hat{k}}{\sqrt{14}}\right)$

$\vec{E} = (-5.15\times10^5)\hat{i} + (1.72\times10^5)\hat{j} + (3.43\times10^5)\hat{k}\ N/c$

2) Electric field p due to q_2

$\hat{r}_2 = \frac{\hat{i}+4\hat{j}-3\hat{k}}{\sqrt{26}}$

$\vec{E}_2 = \frac{9\times10^9(-2\times10^{-3})}{26}\left(\frac{\hat{i}+4\hat{j}-3\hat{k}}{\sqrt{26}}\right)$

$\vec{E}_2 = (-6.92\times10^5)\hat{i} - (2.77\times10^6)\hat{j} + (2.08\times10^6)\hat{k}\ N/c$

$E = \vec{E}_1 + \vec{E}_2 = (-1.21\hat{i} - 2.60\hat{j} + 2.42\hat{k})\times10^6\ N/C$

Now electric force $\vec{F} = q_0\vec{E}$

$\vec{F} = 10\times10^{-9}(-1.21\hat{i} - 2.60\hat{j} + 2.42\hat{k})\times10^6\ N/c$

$\boxed{\vec{F} = (-0.0121\hat{i} - 0.0260\hat{j} + 0.0242\hat{k})}\ N$

2) The finite sheet $0\le x\le 1$, $0\le y\le 1$ on the $z=0$ plane has a charge density $\rho_s = xy(x^2+y^2+25)^{3/2}\ nC/m^2$ Find

- Total charge on sheet
- The eletric field at (0,0,5)

· The force Experienced by $-2\,mC$ charge located at $(0,0,5)$

Sol: Given

limit or point $0 \le x \le 1$, $0 \le y \le 1$, $z = 0$

Surface charge density $\rho_s = xy(x^2+y^2+25)^{3/2}\ nC/m^2$

We know that surface charge

$$Q = \iint \rho_s\, ds$$

$$= \int_0^1\int_0^1 xy(x^2+y^2+25)^{3/2}\,dx\,dy$$

∴ let $u = x^2+y^2+25$

$du = 2x\,dx + 2y\,dy$

$xy\,dx\,dy$

$$Q = \frac{1}{6}\left[(x^2+y^2+25)^{3/2}\right]_{0,0}^{1,1}$$

$$Q = \frac{1}{6}\left[140.3 - 125\right]$$

$$\boxed{Q = 2.55\,nC}$$

b) Electric field at point $(0,0,5)$

points $(x,y,0)$ & $(0,0,5)$ $\vec{R} = -x\hat{i} - y\hat{j} + 5\hat{k}$ $R = \sqrt{x^2+y^2+25}$

We know that

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\int_0^1\int_0^1 \frac{xy(x^2+y^2+25)^{3/2}}{(x^2+y^2+25)^{3/2}}(-x\hat{i} - y\hat{j} + 5\hat{k})\,dx\,dy$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\int_0^1\int_0^1 5xy\,dx\,dy$$

$$\vec{E} = \frac{5}{4}\,\frac{1}{4\pi\epsilon_0}$$

$$\boxed{\vec{E} = 11.25\times10^{9}\,\hat{k}\ N/C}$$

c) Force on $-1\,mC$ charge

We know that $\vec{F} = q\vec{E}$

$$\vec{F} = -1\times10^{-3}\,(11.25\times10^{9})\,\hat{k}\ N/C$$

$$\boxed{\vec{F} = -1.125\times10^{7}\,\hat{k}\ N}$$

3) The point charge $-1nc$, $4nc$, and $3nc$ are located at $(0,0,0)$ $(0,0,1)$ $(1,0,0)$ respectively. Find energy system?

Sol: Given that

$q_1 = -1 \times 10^{-9} c \longrightarrow P_1(0,0,0)$

$q_2 = 4 \times 10^{-9} c \longrightarrow P_2(0,0,1)$

$q_3 = 3 \times 10^{-9} c \longrightarrow P_3(1,0,0)$

We know that $U = \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}} \longrightarrow ①$

$U_{total} = U_{12} + U_{23} + U_{31} \longrightarrow ②$

Distance between q_1 & q_2 $= r_{12} = \sqrt{(0-0)^2 + (0-0)^2 + (1-0)^2} = 1m$

Distance between q_2 & q_3 $= r_{23} = \sqrt{(1-0)^2 + (0-0)^2 + (0-1)^2} = \sqrt{2}m$

Distance between q_3 & q_1 $= r_{31} = 1m$

$U_{12} = 9 \times 10^{9} (-1 \times 10^{-9})(4 \times 10^{-9}) = -36 \times 10^{-9} J$

$U_{23} = \frac{9 \times 10^{-9} (4 \times 10^{-9})(3 \times 10^{-9})}{\sqrt{2}} = 76.4 \times 10^{-9} J$

$U_{31} = -27 \times 10^{-9} J$

$U_{total} = U_{12} + U_{23} + U_{31}$

$= -36 \times 10^{-9} + 76.4 \times 10^{-9} - 27 \times 10^{-9}$

$\boxed{U = 13.4 \times 10^{-9} J}$